



Pepe & Minimum Summands

*Through the dark forests, crossing seven seas and thirteen rivers, Pepe has finally reached the mythical **Island of Bika**. There he found a cave, some old saying says that whoever finds a cave on the Island of Bika and enters it, never comes back alive, but Pepe is known as the most courageous in his village. So, he decides to enter it. There he finds a locked chest and a note beside it. The entrance immediately closes but fortunately Pepe had come with a torch. Pepe starts reading the note:*

You are given two positive integers n, x . There's an array a . Let's consider m is the size of the array a . a_i is the base of i^{th} ($1 \leq i \leq m$) summand of n and x is the power of every summand of n . Formally, $n = a_1^x + a_2^x + \dots + a_m^x$.

Your task is to find out the minimum possible value of m . In other words, the minimum size of a .

Pepe has to enter the answer and the array a in the chest which contains a smart lock system. It is heard that Island of Bika used to be the greatest in technology.

Help Pepe get the treasure and escape the cave.

Input Format

The first line of the input contains a single integer T , number of test cases.

Each test case contains two positive integers n, x .

Output Format

For each test case:

- The first line should contain a single integer m , minimum possible size of a .
- The second line should contain the integers in a ($a_1, a_2, \dots, a_m$) in ascending order.

If you don't output the integers in a and only output m , the minimum possible size of a , then you may obtain half points for the specific subtask.

If there are multiple possible array a output any of them.

Constraints

- $1 \leq T \leq 1000$
- $1 \leq n \leq 10^5$
- $1 \leq x \leq 10$

Subtasks

Subtask	Score	Additional Constraints
1	7	$x = 1$
2	8	$2^x > n$
3	8	$m = 2$
4	10	$x = 2 ; 1 \leq n \leq 100$
5	14	$1 \leq n \leq 1000$
6	25	$1 \leq n \leq 10^5 ; 2 \leq x \leq 3$
7	28	No additional constraints.

Examples

```
4
65 2
10 3
1024 10
12 8
```

The correct output is

```
2
1 8
3
1 1 2
1
2
12
1 1 1 1 1 1 1 1 1 1 1 1 1
```

Another correct output is

```
2
3
1
12
```

But it will result half points.

Explanation

In the first test case, $n = 65$ and $x = 2$. A possible array a is:

$a = [1, 8]$ because, $65 = 1^2 + 8^2$.

In the second test case, $n = 10$ and $x = 3$. A possible array a is:

$a = [1, 1, 2]$ because, $10 = 1^3 + 1^3 + 2^3$.

In the fifth test case, $n = 12$ and $x = 8$. The only possible array a is:

$a = [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]$ because, $12 = 1^8 + 1^8 + 1^8 + 1^8 + 1^8 + 1^8 + 1^8 + 1^8 + 1^8 + 1^8 + 1^8 + 1^8$.